

Unsupervised Learning on Country Data

Luca Cicu
Mat. 100001
l.cicu@campus.unimib.it

Lorenzo Liuzzo
Mat. 942123
l.liuzzo2@campus.unimib.it

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

Exploratory Data Analysis

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

Column Name	Description
country	Name of the country.
child_mort	Death of children under 5 years of age per 1000 live births.
exports	Exports of goods and services per capita. Given as %age of the GDP per capita.
health	Total health spending per capita. Given as %age of GDP per capita.
imports	Imports of goods and services per capita. Given as %age of the GDP per capita.
income	Net income per person.
inflation	The measurement of the annual growth rate of the Total GDP.
life_expec	The average number of years a new born child would live if the current mortality patterns are to remain the same.
total_fer	The number of children that would be born to each woman if the current age-fertility rates remain the same.
gdpp	The GDP per capita. Calculated as the Total GDP divided by the total population.

Table 1: Dataset description

Column Distributions

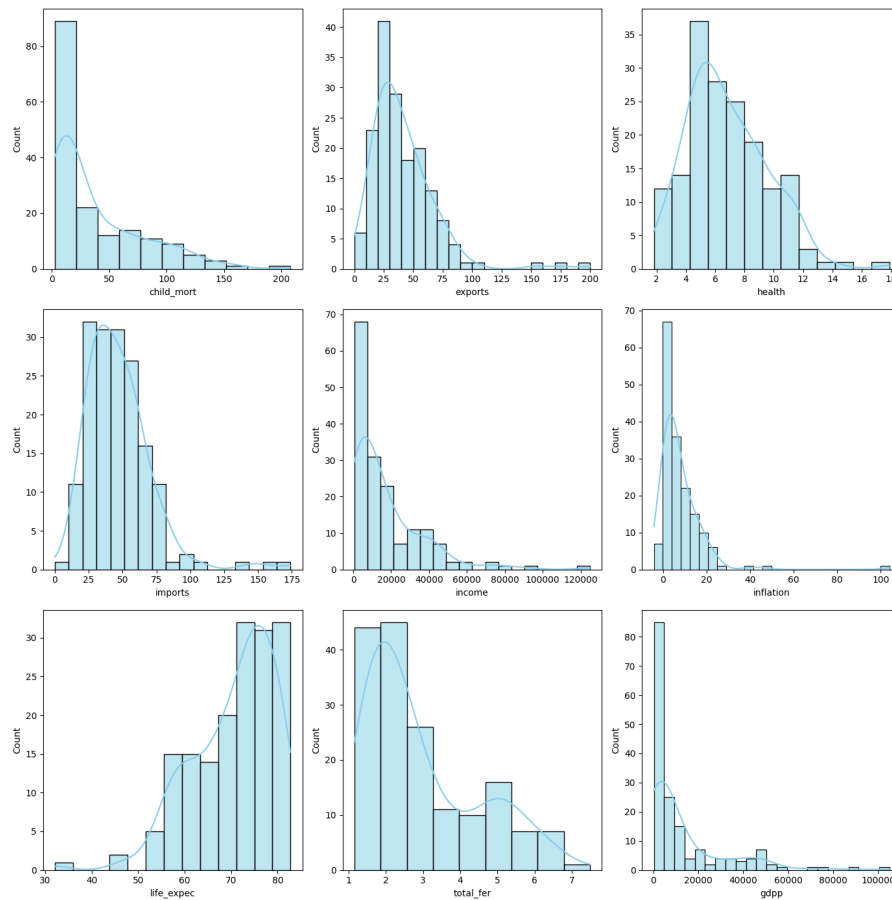


Figure 1: Columns distribution of X and Y, after a transformation.

Correlation Matrix

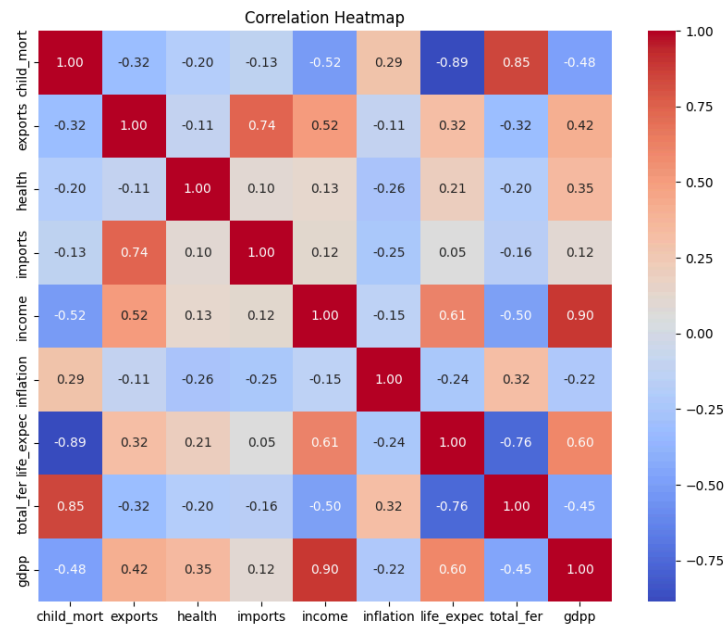


Figure 2: Correlation matrix

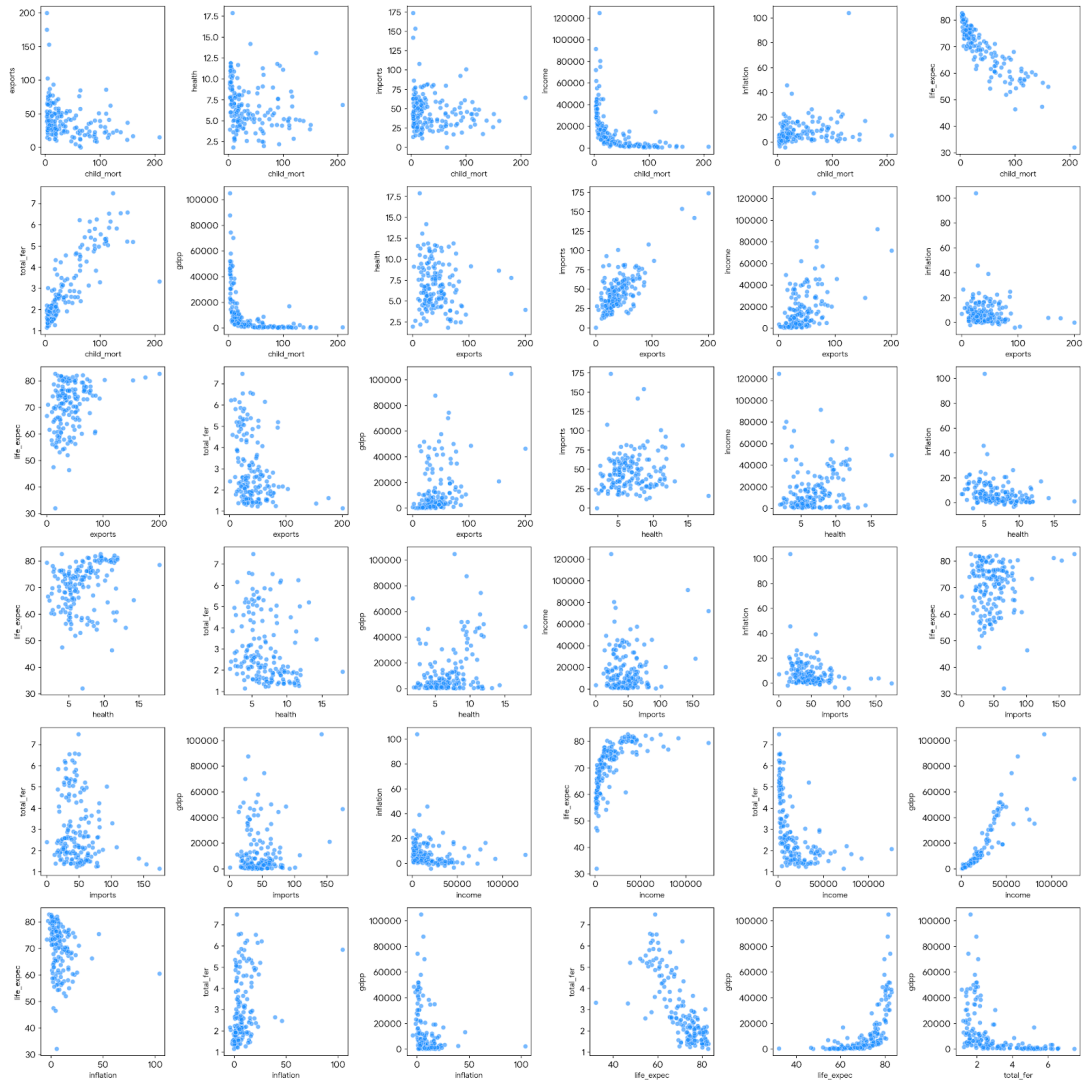


Figure 3: Pairwise scatterplot.

t-SNE

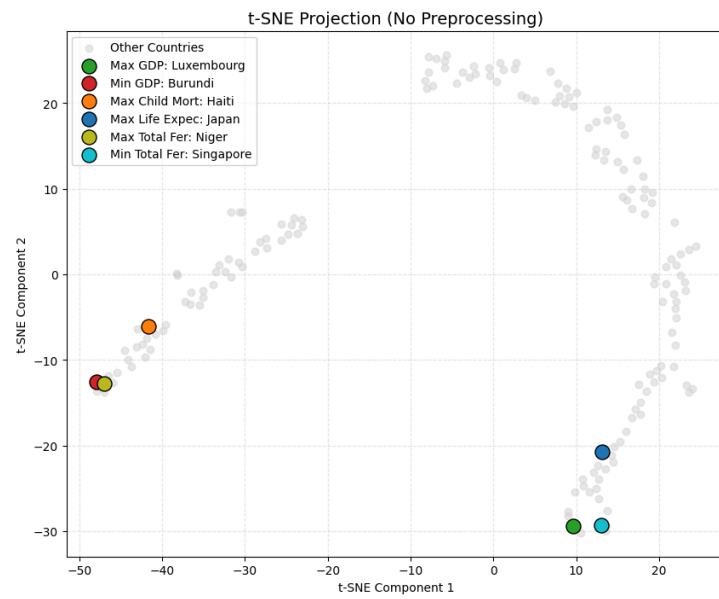


Figure 4:

Preprocessing Strategy

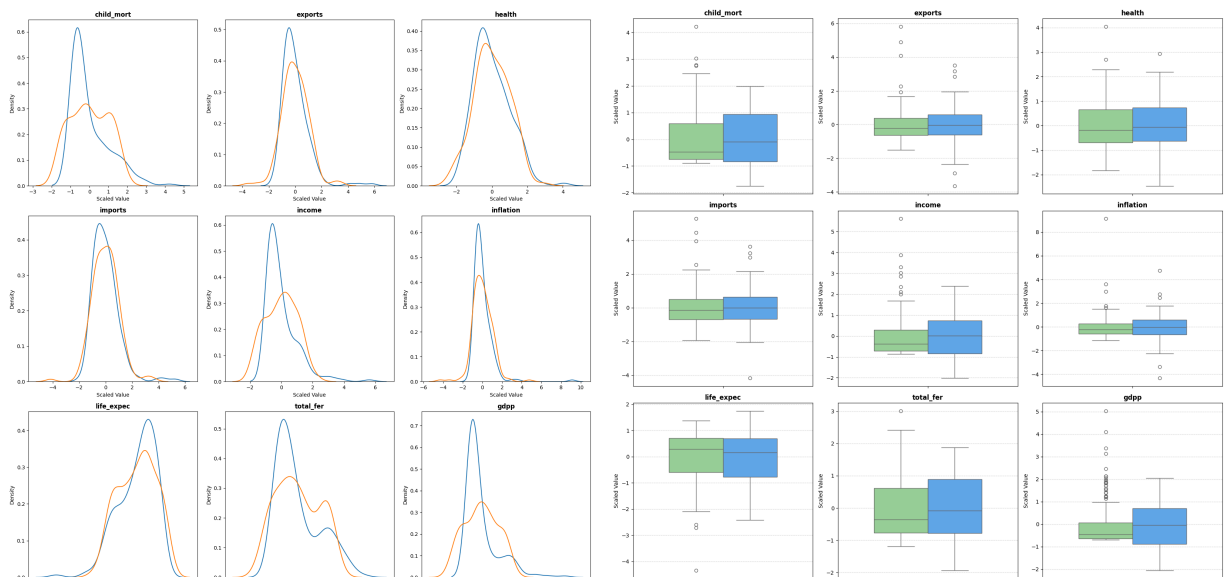


Figure 5: Diocan

Clustering

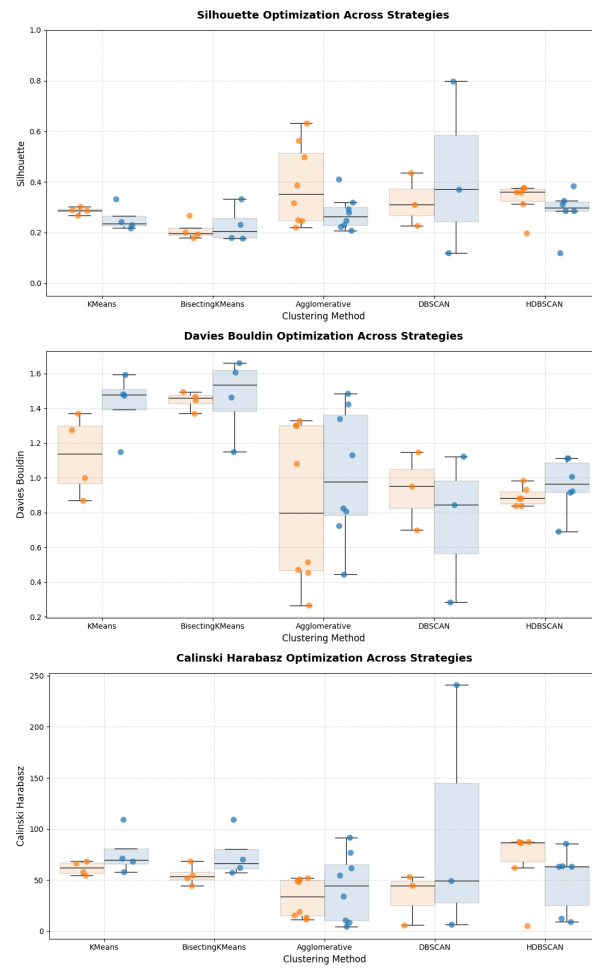


Figure 6: Diocan

Clustering Validation

Bayesian Network

Conditional Independence Tests

PC Algorithm

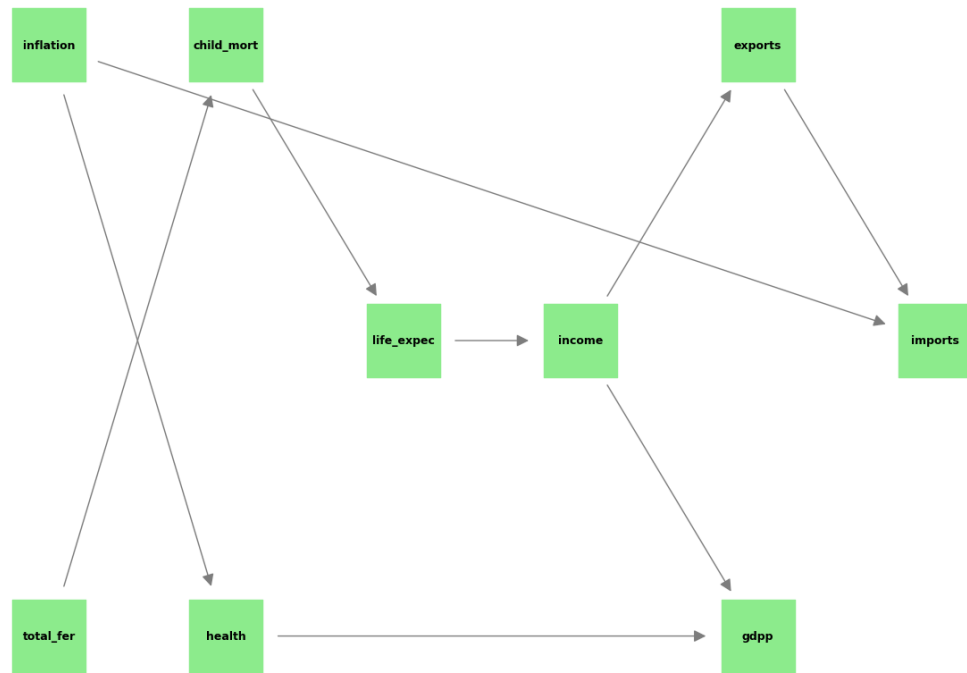


Figure 7: Directed Acyclic Graph builded using PC